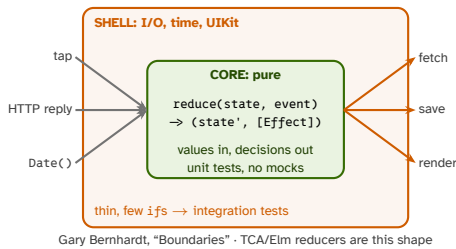


In one line: Push decisions into **pure functions over immutable values**, push effects to a **thin imperative shell**, and use the **type system** (enums, smart constructors) so that **illegal states cannot be written down**. In Swift FP is a strong *default*, not a religion: local mutation of values is fine; the frameworks are object-oriented.

Functional core, imperative shell



Gary Bernhardt, "Boundaries" - TCA/Elm reducers are this shape

Make illegal states unrepresentable

loading	items?	error?	
F	nil	nil	✓
T	nil	nil	✓
F	[x]	nil	✓
F	nil	e	✓
T	[x]	nil	×
T	nil	e	×
F	[x]	e	×
T	[x]	e	×

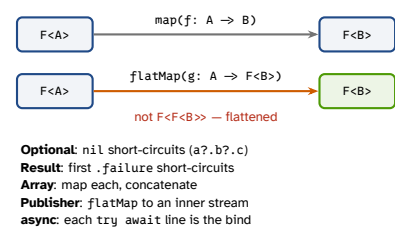
enum Loadable<V>
case idle
case loading
case loaded(V)
case failed(E)

1 + 1 + V + E:
every value legal

2 × 2 × 2 = 8 states, 4 legal

product = multiply (struct) · sum = add (enum)

map / flatMap: one shape



How it works

- Pure function:** same input → same output, **no side effects** (I/O, shared mutation, logging). **Referential transparency:** a call can be replaced by its value. Hidden inputs break it: Date(), random, .shared, globals — pass them in.
- Value semantics:** struct/enum copy on assignment; Array/Dictionary/String are **copy-on-write**. A mutating method on a local value is *not* impure — nobody else can see it. A struct holding a class reference has lost value semantics.
- Algebraic data types:** product (struct, tuple: states multiply) and sum (enum with associated values: states add). Exhaustive switch makes the compiler find every site when a case is added.
- Total function:** defined for every input. Partial ones trap: a[i], first!, Int(d) on .nan. Fix by widening the output (Optional/Result) or narrowing the input (a NonEmpty type).
- Parse, don't validate** (Alexis King): a validator returns Bool and keeps the raw String — every layer re-checks or forgets. A **parser/smart constructor** returns a *more precise type* that is the proof: init?/throws init, then the rest of the code takes Email, not String.
- Higher-order functions** take or return functions: map, filter, reduce, compactMap, key paths as functions (users.map(\.name), Swift 5.2), retry/memoize wrappers. **Composition:** g(f(x)); custom operators cost readability.
- Functor / monad in plain words:** map transforms inside the box and keeps its shape (map(id) changes nothing; two maps = one map of the composition); flatMap chains a step that returns its own box and flattens; a way to put a value in the box (.some, .success, [x], Just) completes the monad.

Example — states and parsing

```
enum Loadable<Value> {
    case idle, loading
    case loaded(Value)
    case failed(any Error)
}
func title(_ s: Loadable<[String]>) -> String { // total
    switch s {
    case .idle:           return "Pull to refresh"
    case .loading:        return "Loading..."
    case .loaded(let xs): return "\(xs.count) items"
    case .failed(let e):  return e.localizedDescription
    }
}
struct Email: Hashable { // smart constructor
    let value: String
    init?(raw: String) {
        let s = raw.lowercased()
        guard s.split(separator: "@").count == 2 else { return nil }
        value = s
    }
}
func invite(_ to: Email) {} // no raw String gets in
```

Dependencies as functions

```
struct UserClient { // a struct of closures
    var fetchName: (Int) async throws -> String
}
extension UserClient {
    static let mock = UserClient(fetchName: { _ in "Ada" })
}
final class ProfileModel {
    let client: UserClient
    init(client: UserClient) { self.client = client }
}
let model = ProfileModel(client: .mock) // no mock class
```

Swap one closure per test — no protocol + mock class (Point-Free “protocol witness” style). Cost: no conformance checking, weaker discoverability, closures may capture self. Partial application pre-binds arguments.

Where FP stops paying in Swift

- Copies:** writing to a *shared* CoW buffer copies all of it — O(n).
- Intermediates:** map.filter.map on Array allocates per step — .lazy or a loop; appending in reduce is O(n²), use reduce(into:).
- No guaranteed tail calls** — deep recursion overflows; loop instead.
- Readability** (operator soup) and **OO frameworks** (UIView, delegates, NSObject) — keep FP in the core.

Traps

- compactMap drops nils; sequence flatMap flattens (split in Swift 4.1, SE-0187); on Optional/Result flatMap is the bind.
- Pure ≠ no var: invisible local mutation is fine; a closure mutating a captured var is not.
- Let on a class reference freezes the pointer, not the object.
- “Loading *with* old items” can be a real state (pull-to-refresh) — model it (case loading(previous: V?)), don't reopen the optionals.

Remember

Values in, decisions out, effects at the edge; enums for “one of”, smart constructors for “checked”.